

Cyber Gadget Gyro Controller Pro

The Cyber Gadget Gyro Controller Pro is a controller that for now, only works with PCs and the Nintendo Switch, however, what makes this controller unique is that you can switch out the positions of the d-pad and the analog joystick. Meaning you can have the two analog joysticks sitting right next to each other like on a PlayStation controller, or you can switch its position with a d-pad and now you've got a layout more akin to the



xbox controller. Switching them out is as simple as popping them out and switching them around. The controller has more up its sleeve, offering motion-sensing controls for Switch games, and plenty of other customisation options, such as button options of height options for the analog joysticks. There's also buttons on the back of the controller, these expansion buttons can be programmed to replicate existing button functions or even to custom-created macros.



Boosting lithium-ion battery capacity

One of the major flaws of rechargeable batteries today is that their capacities degrade pretty quickly. Just after a year of regularly charging a li-ion battery, it can no longer charge at its full capacity anymore. However, this could soon change. A team of scientists working at the Japan Advanced Institute of Science and Technology (JAIST) have been working on trying to maintain li-ion battery capacity for longer durations. Graphite anodes are used in li-ion batteries, and they are paired with a binder material because they can fall apart without it. Li-ion batteries today use poly(vinylidene fluoride) (PVDF) as a binder. However, it was found that with this binder, after just 500 charge-discharge cycles only 65 percent of capacity is usable. In their research, the JAIST came across a new binder material, called bis-imino-acenaphthenequinone-paraphenylene (BP) copolymer. This new binder can supposedly allow the battery to maintain 95 percent capacity for more than 1700 charge-discharge cycles. That means being able to almost fully charge the battery, everyday, for five years.

Apex H2 motorcycle

Segway, the company that's known for its upright scooters from a decade or two ago, have come out with a motorcycle of all things, and it looks really cool. The Apex H2 motorcycle is a hybrid hydrogen-electric motorcycle, and the first from the company which was reportedly acquired by Chinese startup Ninebot in 2015. Since then, the company has shifted to EVs, and has worked on rugged four-wheelers, delivery robots and go-karts, among other things. Segway has been putting out concepts for a while now, but the Apex H2 motorcycle is something they're going to be putting to production. Like we mentioned, this is a hybrid hydrogen-electric bike, meaning it pulls energy both from a battery and hydrogen fuel cells. This allows for much better mileage than traditional EVs. The Apex H2 motorcycle can go from 0 to 100kmh in under 4 seconds, and has a max speed of 150 kmh, which is slower than a lot of motorcycles, but hey, it's faster than a segway, and it emits water vapour from its exhaust.

